

lab_code_14.c

```
1 // Name: Md Shagor
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4 // Lab Report: 14
5 // Experiment Name: Linear Search algorithm
6
7 #include <stdio.h>
8
9 int main() {
10     // Increased safety: Check the array bounds
11     int arr[100], n, key, i, found = 0;
12
13     printf("Enter number of elements (Max 100): ");
14     scanf("%d", &n);
15
16     // Fix 1: Basic validation to prevent memory issues
17     if (n > 100 || n <= 0) {
18         printf("Invalid size! Please enter a number between 1 and 100.\n");
19         return 1;
20     }
21
22     printf("Enter %d array elements:\n", n);
23     for (i = 0; i < n; i++) {
24         scanf("%d", &arr[i]);
25     }
26
27     printf("Enter element to search: ");
28     scanf("%d", &key);
29
30     // Linear Search Logic
31     for (i = 0; i < n; i++) {
32         if (arr[i] == key) {
33             // Fix 2: Added \n for better readability
34             printf("Element found at position %d\n", i + 1);
35             found = 1;
36             break; // Stop searching once found
37         }
38     }
39
40     if (found == 0) {
41         printf("Element not found in the array.\n");
42     }
43
44     return 0;
45 }
46
```